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298-299页2,4,5,6,7,10,12.

290-2

Apply the Gram-Schmidt process to the vectors $\beta_1 = (1, 0, 1)$, $\beta_2 = (1, 0, -1)$, $\beta_3 = (0, 3, 4)$, to obtain an orthonormal basis for $\mathbb{R}^3$ with standard inner product.

Solution: $e_1 = \beta_1 / \|\beta_1\| = \frac{1}{\sqrt{2}}(1, 0, 1)$.

$\alpha_2 = \beta_2 - (\beta_2 \cdot e_1)e_1 = \beta_2$, $e_2 = \alpha_2 / \|\alpha_2\| = \frac{1}{\sqrt{2}}(1, 0, -1)$.

$\alpha_3 = \beta_3 - (\beta_3 \cdot e_1)e_1 - (\beta_3 \cdot e_2)e_2 = (0, 3, 0)$, $e_3 = (0, 1, 0)$.

290-3

Consider $\mathbb{C}^3$, with the standard inner product. Find an orthonormal basis for the subspace spanned by $\beta_1 = (1, 0, i)$ and $\beta_2 = (2, 1, 1 + i)$.

Solution: $e_1 = \left(\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}i}{2}\right)$, $e_2 = \left(\frac{1-i}{4}, \frac{1}{2}, \frac{3-i}{4}\right)$.

290-6

Let W be a subspace of $\mathbb{R}^2$ spanned by $(3, 4)$. Using the standard inner product, let E be the orthogonal projection of $\mathbb{R}^2 \rightarrow W$. Find

(a) $E(x_1, x_2)$; (b) the matrix of E in standard basis; (c) $W^\perp$; (d) an orthonormal basis in which E is the matrix $\text{diag}(1, 0)$.

Solution: (a) $E(x_1, x_2) = \frac{1}{25}(3x_1 + 4x_2)(3, 4) = \frac{1}{25}(9x_1 + 12x_2, 12x_1 + 16x_2)$.

(b) $E = \begin{pmatrix} \frac{9}{25} & \frac{12}{25} \\ \frac{12}{25} & \frac{16}{25} \end{pmatrix}$.

(c) $W^\perp = \langle (4, -3) \rangle$. (d) $e_1 = \left(\frac{3}{5}, \frac{4}{5}\right)$, $e_2 = \left(\frac{4}{5}, -\frac{3}{5}\right)$.

290-10

Let $V = \mathbb{C}^{n \times n}$, with inner product $\langle A, B \rangle = \text{tr}(AB^*)$. Find the orthogonal complement of diagonal matrices.

Solution: It is clearly the matrices whose diagonal is zero.

290-12

Let W be a finite-dimensional subspace of an inner product space V , and E be the orthogonal projection $V \rightarrow W$. Prove that $\langle E\alpha, \beta \rangle = \langle \alpha, E\beta \rangle$ for all $\alpha, \beta \in V$.

Proof: $\langle \alpha - E\alpha, E\beta \rangle = 0$ implies $\langle \alpha, E\beta \rangle = \langle E\alpha, E\beta \rangle$ and likewise $\langle E\alpha, \beta \rangle = \langle E\alpha, E\beta \rangle$, so $\langle \alpha, E\beta \rangle = \langle E\alpha, \beta \rangle$.

290-13

Let $S \subset V$, show that $(S^\perp)^\perp$ contains $\text{Span}(S)$. When V is finite-dimensional, show that $(S^\perp)^\perp = \text{Span}(S)$.

Proof: Clearly $S \subset (S^\perp)^\perp$ and $(S^\perp)^\perp$ is a subspace, so $\text{Span}(S) \subset (S^\perp)^\perp$. If V is finite dimensional, then $\dim(S^\perp)^\perp = \dim V - \dim(S^\perp) = \dim \text{Span}(S)$, so $\text{Span}(S) = (S^\perp)^\perp$.

290-15

Suppose $V = W_1 \oplus W_2$ and f_1, f_2 are inner products on W_1, W_2 . Show that there is a unique inner product f on V such that (a) $W_2 = W_1^\perp$; (b) $f(\alpha, \beta) = f_k(\alpha, \beta)$ when α, β are in W_k , $k = 1, 2$.

Proof: For any $\alpha = \alpha_1 + \alpha_2$ and $\beta = \beta_1 + \beta_2$, define $\langle \alpha, \beta \rangle = f_1(\alpha_1, \beta_1) + f_2(\alpha_2, \beta_2)$. This is clearly an inner product on V . If $\alpha \in W_1$ and $\beta \in W_2$ then $\langle \alpha, \beta \rangle = 0$, combined with $W_1 + W_2 = V$ we have $W_2 = W_1^\perp$. By definition $\langle \alpha, \beta \rangle = f_k(\alpha, \beta)$ if $\alpha, \beta \in W_k$.

290-16

Let W be a finite-dimensional subspace of V . There are many projections with W as their range. One of these, the orthogonal projection on W , has the property that $\|E\alpha\| \leq \|\alpha\|$ for every $\alpha \in V$. Prove that if E is a projection with range W , such that $\|E\alpha\| \leq \|\alpha\|$ then E is the orthogonal projection on W .

Proof: Since E is a projection, $V = W \oplus \text{Ker} E$, we show that $\text{Ker} E \perp W$. For any $\alpha \in \text{Ker} E \setminus \{0\}$ and $\beta \in W$, then $\|E(t\alpha + \beta)\| \leq \|t\alpha + \beta\|$ implies $t^2 \langle \alpha, \alpha \rangle + 2\text{Re}(t \langle \alpha, \beta \rangle) \geq 0$. Let $t = -\langle \beta, \alpha \rangle / \|\alpha\|^2$ we obtain $\langle \alpha, \beta \rangle = 0$.

298-2

Let $T \in \mathcal{L}(\mathbb{C}^2)$ defined by $Te_1 = (1 + i, 2)$, $Te_2 = (i, i)$. Find the matrix of T^* , does T commute with T^* ?

Solution: $T^* = \begin{pmatrix} 1-i & -i \\ 2 & -i \end{pmatrix}$. They do not commute.

298-4

Let V be a finite dim inner product space and $T \in \mathcal{L}(V)$. Show that $\text{Im} T^* = (\text{Ker} T)^\perp$.

Proof: Clearly $\text{Im} T^* \subset (\text{Ker} T)^\perp$: If $\beta = T^* \gamma \in \text{Im} T^*$ and $\alpha \in \text{Ker} T$, then $\langle \alpha, \beta \rangle = \langle \alpha, T^* \gamma \rangle = \langle T\alpha, \gamma \rangle = 0$. Since $T^{**} = T$, $\text{Im} T \subset (\text{Ker} T^*)^\perp$, and $\dim \text{Im} T^* + \dim \text{Ker} T^* = \dim \text{Im} T + \dim \text{Ker} T = \dim V$, so $\dim \text{Im} T^* = \dim \text{Im} T = \dim (\text{Ker} T)^\perp = \dim (\text{Ker} T^*)^\perp$, and $\text{Im} T^* = (\text{Ker} T)^\perp$.

298-5

If $T \in \mathcal{L}(V)$ is invertible, then $(T^*)^{-1} = (T^{-1})^*$.

Proof: $1 = (TT^{-1})^* = (T^{-1})^* T^*$ so $(T^*)^{-1} = (T^{-1})^*$.

298-6

Let $\beta, \gamma \in V$. Show that $T\alpha = \langle \alpha, \beta \rangle \gamma$ defines a linear operator on V . Show that T has an adjoint, and describe T^* explicitly.

Proof: $T \in \mathcal{L}(V)$ is trivial. Its adjoint is $T^* \alpha = \langle \alpha, \gamma \rangle \beta$, since $\langle T\alpha, \alpha' \rangle = \langle \alpha, \beta \rangle \cdot \langle \gamma, \alpha' \rangle = \langle \alpha, T^* \alpha' \rangle$.

298-7

Show that the product of two self-adjoint operators is self-adjoint iff they commute.

Proof: $(TU)^* = U^* T^* = UT$ so $(TU)^* = TU$ iff $TU = UT$.

298-10

Let $V = \mathbb{C}^{n \times n}$ with the inner product $\langle A, B \rangle = \text{tr}(AB^*)$. Let $P \in \text{GL}(V)$ and $T_P \in \mathcal{L}(V)$, $A \mapsto P^{-1}AP$. Find the adjoint of T_P .

Solution: $(T_P)^* : A \mapsto (P^{-1})^* A P^*$ since

$$\langle T_P A, B \rangle = \text{tr}(P^{-1} A P B^*) = \text{tr}(A P B^* P^{-1}) = \langle A, T_P^* B \rangle.$$

298-12

Let V be a finite dim complex vector space, $T \in \mathcal{L}(V)$. Prove that T is self-adjoint iff $\langle T\alpha, \alpha \rangle \in \mathbb{R} \forall \alpha \in V$.

Prof: If $T = T^*$ then $\langle T\alpha, \alpha \rangle = \langle \alpha, T^* \alpha \rangle = \langle \alpha, T\alpha \rangle = \overline{\langle T\alpha, \alpha \rangle}$ so $\langle T\alpha, \alpha \rangle \in \mathbb{R} \forall \alpha \in V$.

Conversely, let $S = T - T^*$ then $\langle S\alpha, \alpha \rangle = \langle T\alpha, \alpha \rangle - \langle \alpha, T\alpha \rangle = 0$. By expanding $\langle S(\alpha + \beta), (\alpha + \beta) \rangle$ we obtain $\langle S\alpha, \beta \rangle + \langle S\beta, \alpha \rangle = 0$. Let $\alpha \mapsto i\alpha$ we obtain $\langle S\alpha, \beta \rangle - \langle S\beta, \alpha \rangle = 0$ hence $\langle S\alpha, \beta \rangle = 0 \forall \alpha, \beta$. Therefore $S = 0$ and $T = T^*$ is self-adjoint.